

Week 6 recitation

Fishery (a): sampling

Fifteen observations were generated from each distribution. The discrete probabilities are 0.20, 0.10, 0.50, 0.10, and 0.10 for outcomes 0, 1, 2, 3, and 4. Independent uniform streams are stored in columns N:S and transformed with Excel formulas.

U(0,1)	Discrete	Bin(15,0.7)	U(5,9)	N(5,2)	Exp(0.4)
0.7311	2	8	6.2428	3.9296	1.7510
0.8357	0	13	8.6414	2.6228	1.7157
0.0275	2	11	7.7720	2.8757	1.8709
0.0704	0	11	7.7356	7.9012	3.3823
0.1045	0	12	5.4996	5.3155	1.6168
0.4317	2	10	7.1175	5.4499	8.1033
0.0486	0	7	6.2982	4.2319	0.6278
0.7888	2	10	8.9414	3.6067	1.9448
0.2247	2	10	6.6190	7.5360	0.1689
0.2537	2	9	8.0868	6.7551	2.1376
0.2683	0	10	7.0744	4.6115	1.3474
0.6082	4	11	5.7140	6.2452	7.9281
0.1148	1	12	6.7303	6.6309	0.6419
0.1398	0	11	6.5819	4.9439	3.4729
0.9024	4	12	8.2108	3.3690	1.4469

Excel capture: Sampling!G2:L17. All 15 observations are shown. N(5,2) uses standard deviation 2; Exp(0.4) uses rate 0.4, with mean 2.5.

Distribution	Excel formula, first observation
Uniform (0,1)	=N3
Discrete	=LOOKUP(O3,\$E\$3:\$E\$7,\$B\$3:\$B\$7)
Binomial (15,0.7)	=BINOM.INV(15,0.7,P3)
Uniform (5,9)	=5+4*Q3
Normal (5,2)	=NORM.INV(R3,5,2)
Exponential (rate 0.4)	=-LN(1-S3)/0.4

Uniform draws were generated with NumPy seed 76006 and frozen as workbook inputs. Transformations, profits, summaries, and confidence limits are Excel formulas. Captures come from the completed Excel workbook; Solver images show the actual dialogs.

Reference: RS W6.pdf, RS W6 Sol.pdf and RS W6 Sol.xlsx were used to check the intended model and workbook structure. Simulation draws and statistics are our own realization.

Fishery (b): 70-day simulation

Daily price is Normal(\$3.65, \$0.20). Demand is sampled independently of price. Sold = MIN(3500,Demand);
Revenue = Price x Sold; Profit = Revenue - \$10,000.

Days	Mean profit	Sample SD	95% CI for mean
70	\$2,084.20	\$1,837.20	\$1,646.14 to \$2,522.27

Mean = AVERAGE(O4:O73), SD = STDEV.S(O4:O73), and CI = mean +/- T.INV.2T(0.05,69) x SD / SQRT(70). The interval estimates expected daily profit.

Day	Price RND	Price	Demand RND	Demand	Sold	Revenues	Cost	Profit
1	0.7311	3.7732	0.6506	5000	3500	\$13,206.34	\$10,000.00	\$3,206.34
2	0.1253	3.4202	0.3107	4000	3500	\$11,970.75	\$10,000.00	\$1,970.75
3	0.2963	3.5430	0.5036	4000	3500	\$12,400.36	\$10,000.00	\$2,400.36
4	0.8357	3.8454	0.1287	3000	3000	\$11,536.20	\$10,000.00	\$1,536.20
5	0.8947	3.9004	0.9103	6000	3500	\$13,651.35	\$10,000.00	\$3,651.35
6	0.1173	3.4123	0.4966	4000	3500	\$11,942.97	\$10,000.00	\$1,942.97
7	0.0275	3.2661	0.7962	5000	3500	\$11,431.41	\$10,000.00	\$1,431.41
8	0.6635	3.7344	0.6930	5000	3500	\$13,070.37	\$10,000.00	\$3,070.37
9	0.1441	3.4376	0.5269	5000	3500	\$12,031.51	\$10,000.00	\$2,031.51
10	0.0704	3.3554	0.0549	2000	2000	\$6,710.74	\$10,000.00	(\$3,289.26)
11	0.6759	3.7413	0.6839	5000	3500	\$13,094.46	\$10,000.00	\$3,094.46
12	0.9266	3.9401	0.7415	5000	3500	\$13,790.42	\$10,000.00	\$3,790.42
13	0.1045	3.3987	0.0465	1000	1000	\$3,398.75	\$10,000.00	(\$6,601.25)
14	0.7446	3.7815	0.1249	3000	3000	\$11,344.49	\$10,000.00	\$1,344.49
15	0.5627	3.6815	0.4762	4000	3500	\$12,885.41	\$10,000.00	\$2,885.41
16	0.4317	3.6156	0.5053	4000	3500	\$12,654.65	\$10,000.00	\$2,654.65
17	0.3525	3.5743	0.5294	5000	3500	\$12,509.92	\$10,000.00	\$2,509.92
18	0.5890	3.6950	0.9609	6000	3500	\$12,932.45	\$10,000.00	\$2,932.45
19	0.0486	3.3182	0.0764	2000	2000	\$6,636.40	\$10,000.00	(\$3,363.60)
20	0.0231	3.2514	0.3246	4000	3500	\$11,380.02	\$10,000.00	\$1,380.02
21	0.3505	3.5732	0.2221	4000	3500	\$12,506.18	\$10,000.00	\$2,506.18
22	0.7888	3.8105	0.5604	5000	3500	\$13,336.63	\$10,000.00	\$3,336.63
23	0.3127	3.5524	0.9853	6000	3500	\$12,433.25	\$10,000.00	\$2,433.25
24	0.2430	3.5107	0.5406	5000	3500	\$12,287.36	\$10,000.00	\$2,287.36
25	0.2247	3.4987	0.5298	5000	3500	\$12,245.46	\$10,000.00	\$2,245.46
26	0.4145	3.6068	0.4047	4000	3500	\$12,623.82	\$10,000.00	\$2,623.82
27	0.8976	3.9036	0.0653	2000	2000	\$7,807.20	\$10,000.00	(\$2,192.80)
28	0.2537	3.5174	0.3092	4000	3500	\$12,310.91	\$10,000.00	\$2,310.91
29	0.2164	3.4931	0.7717	5000	3500	\$12,225.85	\$10,000.00	\$2,225.85
30	0.8099	3.8255	0.5747	5000	3500	\$13,389.28	\$10,000.00	\$3,389.28
31	0.2683	3.5264	0.1877	4000	3500	\$12,342.44	\$10,000.00	\$2,342.44
32	0.3503	3.5731	0.5186	5000	3500	\$12,505.89	\$10,000.00	\$2,505.89
33	0.4230	3.6112	0.4167	4000	3500	\$12,639.03	\$10,000.00	\$2,639.03
34	0.6082	3.7049	0.9703	6000	3500	\$12,967.26	\$10,000.00	\$2,967.26
35	0.6362	3.7197	0.1785	3000	3000	\$11,159.02	\$10,000.00	\$1,159.02

Excel capture: Fishery operations!G3:O38. Days 1-35; the remaining 35 days appear on the next page.

Fishery (b): days 36-70

36	0.7332	3.7745	0.9580	6000	3500	\$13,210.82	\$10,000.00	\$3,210.82
37	0.1148	3.4098	0.2906	4000	3500	\$11,934.19	\$10,000.00	\$1,934.19
38	0.7470	3.7830	0.4326	4000	3500	\$13,240.52	\$10,000.00	\$3,240.52
39	0.7926	3.8131	0.2264	4000	3500	\$13,345.82	\$10,000.00	\$3,345.82
40	0.1398	3.4338	0.1451	3000	3000	\$10,301.37	\$10,000.00	\$301.37
41	0.5881	3.6945	0.3955	4000	3500	\$12,930.80	\$10,000.00	\$2,930.80
42	0.4888	3.6444	0.7507	5000	3500	\$12,755.37	\$10,000.00	\$2,755.37
43	0.9024	3.9090	0.9212	6000	3500	\$13,681.55	\$10,000.00	\$3,681.55
44	0.8497	3.8571	0.8027	6000	3500	\$13,499.68	\$10,000.00	\$3,499.68
45	0.2074	3.4869	0.4394	4000	3500	\$12,204.15	\$10,000.00	\$2,204.15
46	0.0787	3.3673	0.1725	3000	3000	\$10,101.79	\$10,000.00	\$101.79
47	0.8991	3.9053	0.5691	5000	3500	\$13,668.63	\$10,000.00	\$3,668.63
48	0.1184	3.4133	0.5420	5000	3500	\$11,946.71	\$10,000.00	\$1,946.71
49	0.1115	3.4063	0.7931	5000	3500	\$11,922.05	\$10,000.00	\$1,922.05
50	0.3498	3.5728	0.6664	5000	3500	\$12,504.83	\$10,000.00	\$2,504.83
51	0.2300	3.5023	0.4376	4000	3500	\$12,257.90	\$10,000.00	\$2,257.90
52	0.2703	3.5276	0.3942	4000	3500	\$12,346.70	\$10,000.00	\$2,346.70
53	0.3712	3.5843	0.0849	2000	2000	\$7,168.50	\$10,000.00	(\$2,831.50)
54	0.9113	3.9198	0.7106	5000	3500	\$13,719.30	\$10,000.00	\$3,719.30
55	0.5854	3.6931	0.4822	4000	3500	\$12,925.99	\$10,000.00	\$2,925.99
56	0.2261	3.4996	0.9015	6000	3500	\$12,248.72	\$10,000.00	\$2,248.72
57	0.8355	3.8452	0.4952	4000	3500	\$13,458.23	\$10,000.00	\$3,458.23
58	0.3852	3.5917	0.4219	4000	3500	\$12,570.78	\$10,000.00	\$2,570.78
59	0.6295	3.7161	0.4844	4000	3500	\$13,006.33	\$10,000.00	\$3,006.33
60	0.6477	3.7258	0.6038	5000	3500	\$13,040.32	\$10,000.00	\$3,040.32
61	0.4609	3.6303	0.1396	3000	3000	\$10,891.03	\$10,000.00	\$891.03
62	0.5488	3.6745	0.8570	6000	3500	\$12,860.81	\$10,000.00	\$2,860.81
63	0.4144	3.6067	0.5433	5000	3500	\$12,623.56	\$10,000.00	\$2,623.56
64	0.7294	3.7722	0.4898	4000	3500	\$13,202.65	\$10,000.00	\$3,202.65
65	0.1698	3.4590	0.5178	5000	3500	\$12,106.42	\$10,000.00	\$2,106.42
66	0.6558	3.7302	0.3892	4000	3500	\$13,055.81	\$10,000.00	\$3,055.81
67	0.0712	3.3567	0.2093	4000	3500	\$11,748.41	\$10,000.00	\$1,748.41
68	0.6573	3.7310	0.7491	5000	3500	\$13,058.62	\$10,000.00	\$3,058.62
69	0.9646	4.0114	0.5307	5000	3500	\$14,039.98	\$10,000.00	\$4,039.98
70	0.5048	3.6524	0.1761	3000	3000	\$10,957.22	\$10,000.00	\$957.22

Excel capture: Fishery operations!G39:O73. Columns match the preceding page: Day, Price RND, Price, Demand RND, Demand, Sold, Revenues, Cost, Profit.

Demand distribution				Mean	
x	Prob.	CDF	Lower	\$2,084.20	
0	0.02	0.02	0	Standard deviation	
1,000	0.03	0.05	0.02	\$1,837.20	
2,000	0.05	0.1	0.05	CI	
3,000	0.08	0.18	0.1	Lower 95% CI	
4,000	0.33	0.51	0.18	\$1,646.14	
5,000	0.29	0.8	0.51	Upper 95% CI	
6,000	0.2	1	0.8	\$2,522.27	

Fishery operations!B2:E10.

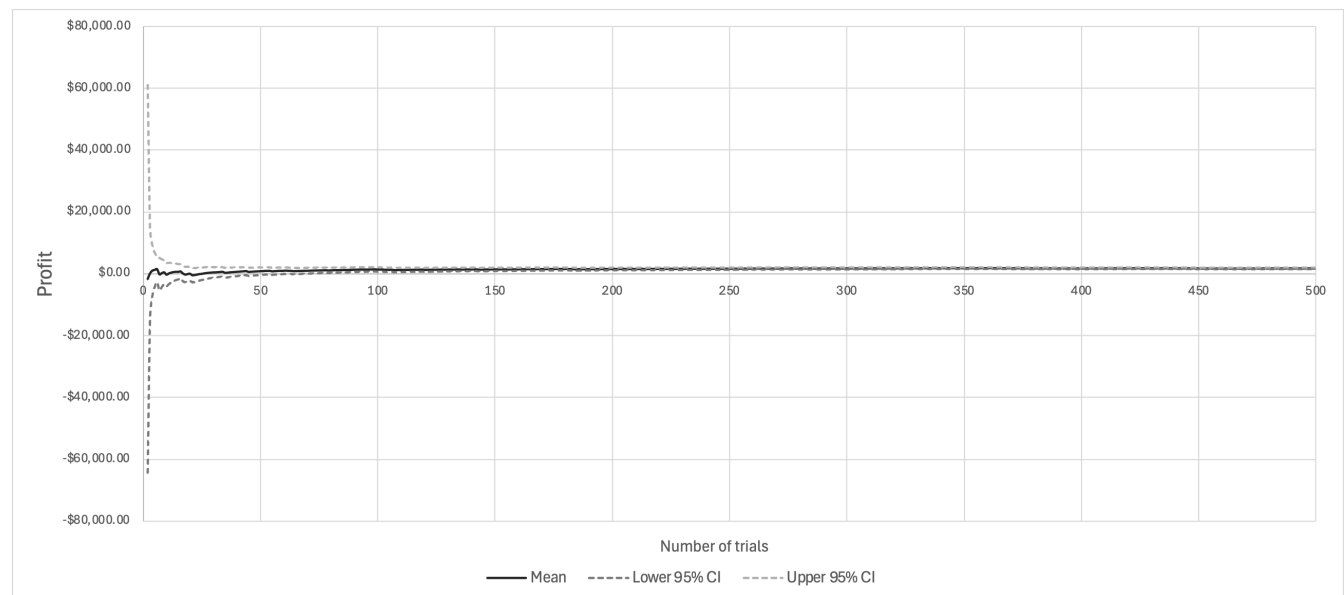
Fishery operations!X4:Y9.

Exact validation: $E[\text{sold}] = 0.02(0) + 0.03(1000) + 0.05(2000) + 0.08(3000) + 0.82(3500) = 3,240$ fish. $E[\text{profit}] = 3.65 \times 3,240 - 10,000 = \mathbf{\$1,826.00}$, inside the simulation confidence interval.

Fishery (c): increasing the sample size

Five hundred days were simulated. Each row uses day 1 through its own day n. Mean = AVERAGE(\$O\$4:Or), SD = STDEV.S(\$O\$4:Or), and CI = mean +/- T.INV.2T(0.05,n-1) x SD / SQRT(n). At n = 1, SD and CI are undefined and left blank.

Days	Mean profit	Sample SD	95% CI for mean
500	\$1,822.20	\$2,762.56	\$1,579.46 to \$2,064.93



Native Excel chart from N analysis. The very wide initial interval reflects the small sample size. All 500 days and cumulative calculations are included in the workbook.

Final cumulative results

491	\$1,809.89	\$2,785.09	\$1,562.94	\$2,056.85
492	\$1,810.94	\$2,782.35	\$1,564.48	\$2,057.40
493	\$1,811.65	\$2,779.56	\$1,565.68	\$2,057.61
494	\$1,813.65	\$2,777.10	\$1,568.16	\$2,059.15
495	\$1,816.56	\$2,775.04	\$1,571.50	\$2,061.63
496	\$1,816.84	\$2,772.25	\$1,572.27	\$2,061.41
497	\$1,817.63	\$2,769.51	\$1,573.55	\$2,061.71
498	\$1,821.44	\$2,768.02	\$1,577.74	\$2,065.14
499	\$1,822.43	\$2,765.33	\$1,579.20	\$2,065.65
500	\$1,822.20	\$2,762.56	\$1,579.46	\$2,064.93

N analysis!P494:T503, days 491-500. Columns: n, Mean, Sample SD, Lower 95% CI, Upper 95% CI.

The running mean need not converge monotonically, but its uncertainty decreases as n grows. The final mean of \$1,822.20 is close to the exact expectation of \$1,826.00. The intervals describe the mean, not daily-profit percentiles.

Fishery (d): choosing the quantity Q

Use the quantity-dependent cost in the solution workbook: $\text{Cost}(Q) = \$3,000 + \$2Q$. Profit = Price x MIN(Q,Demand) - Cost(Q). A native one-input Excel Data Table evaluates Q = 2,000 to 6,000 in steps of 100, using the same 400 draws for every Q.

Fishing	\$
2000	-
2100	\$33.78
2200	\$92.04
2300	\$217.86
2400	\$343.68
2500	\$469.50
2600	\$595.32
2700	\$721.14
2800	\$846.96
2900	\$972.77
3000	\$1,098.59
3100	\$1,224.41
3200	\$1,322.45
3300	\$1,420.50
3400	\$1,518.54
3500	\$1,616.58
3600	\$1,714.62
3700	\$1,812.66
3800	\$1,910.70
3900	\$2,008.75
4000	\$2,106.79
4100	\$2,204.83
4200	\$2,186.01
4300	\$2,167.20
4400	\$2,148.38
4500	\$2,129.57
4600	\$2,110.75
4700	\$2,091.94
4800	\$2,073.12
4900	\$2,054.31
5000	\$2,035.49
5100	\$2,016.68
5200	\$1,997.86
5300	\$1,979.04
5400	\$1,960.22
5500	\$1,941.40
5600	\$1,922.58
5700	\$1,903.76
5800	\$1,884.94
5900	\$1,866.12
6000	\$1,847.30

Inputs and summary

INPUTS

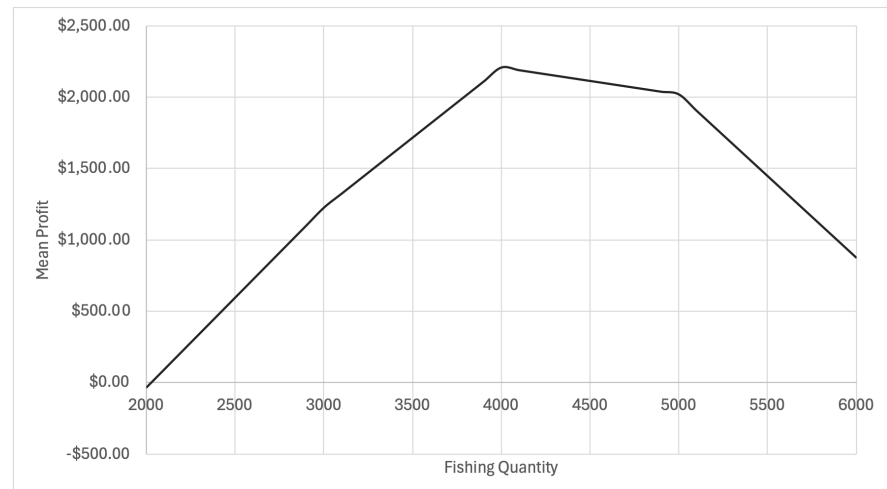
Q (fishes) 4000

OUTPUTS

Mean profit \$2,204.83
 Standard dev \$3,343.04
 Lower 95% CI \$1,876.22
 Upper 95% CI \$2,533.44
 n 400

Fixed cost 3000
 Cost per fish 2

Profit as a function of Q



Recommendation: Q = 4,000 fish. Simulated mean profit is \$2,204.83. Exact expected profit is $3.65 \times 3,650 - (3,000 + 2 \times 4,000) = \$2,322.50$.

Beyond 4,000, marginal expected revenue falls below the \$2 cost per additional fish.

T3:U44: native Data Table.

Excel captures: Analysis on Q!Q3:R14, T3:U44, and its chart. Output U3 = R7; input column T4:T44; column input cell R4. R7 averages O4:O403, so n = 400.

Piedmont airline (a) and (b)

(a) Demand ranges from 14 to 25. Tickets sold $B = \text{MIN}(22, \text{Demand})$. No-shows follow $\text{Binomial}(B, 0.1)$; show-ups = $B - \text{no-shows}$. Profit = $\$150 \times B - \$500 \times \text{MAX}(\text{show-ups} - 19, 0)$.

The 100-flight simulation gives mean **\$2,749.50**, sample SD **\$362.27**, and 95% CI **\$2,677.62 to \$2,821.38**. Exact expected profit is \$2,733.17. Different random draws give different valid simulation statistics.

The interval uses Student t with 99 degrees of freedom and standard error $\text{SD} / \text{SQRT}(100)$.

Number of seats	22
Ticket price	150
Excess cost	500
Cancel. Prob.	0.1
Plane capacity	19

Summary

n	100
Mean	\$2,749.50
Sample SD	\$362.27
Lower 95% CI	\$2,677.62
Upper 95% CI	\$2,821.38

Airline (a)!R2:S13.

Sample	Draw	Demand	Sold	Revenue	Draw	No-shows	Bumped	Penalty	Profit
1	0.8774	22	22	3300	0.5866	2	1	500	2800
2	0.7461	21	21	3150	0.4504	2	0	0	3150
3	0.2271	17	17	2550	0.1699	1	0	0	2550
4	0.9647	24	22	3300	0.6731	3	0	0	3300
5	0.5505	20	20	3000	0.2863	1	0	0	3000
6	0.2070	17	17	2550	0.4679	1	0	0	2550
7	0.4915	19	19	2850	0.1579	1	0	0	2850
8	0.5750	20	20	3000	0.2230	1	0	0	3000
9	0.8103	21	21	3150	0.8114	3	0	0	3150
10	0.3283	18	18	2700	0.3066	1	0	0	2700
11	0.9641	24	22	3300	0.5229	2	1	500	2800
12	0.7845	21	21	3150	0.1537	1	1	500	2650
13	0.1404	16	16	2400	0.5320	2	0	0	2400

Airline (a)!B3:K16. The workbook contains all 100 flights; the capture shows the model and first 13 observations.

(b) One hundred samples give a rough estimate, but the observed confidence half-width is \$71.88. To target a \$10 half-width, the pilot SD gives $n \geq (1.96 \times 362.27 / 10)^2$, rounded up to 5,042. Recommend **5,100 flights**. This targets greater precision than the solution guide's approximate 300-flight recommendation.

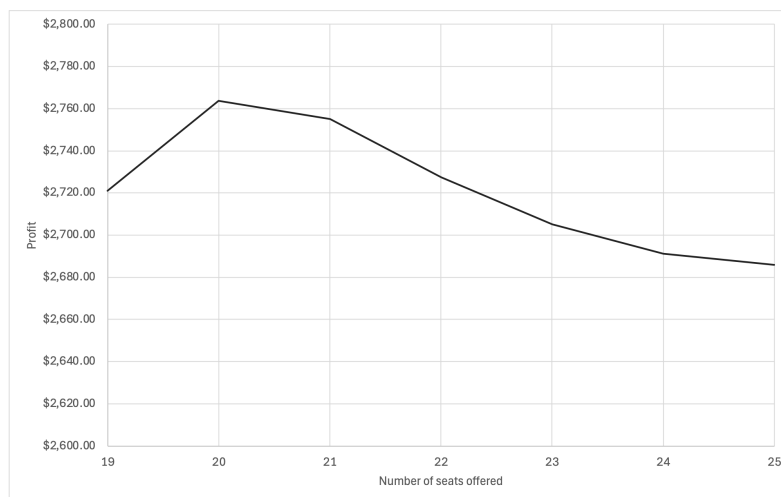
The 5,100-flight run gives mean \$2,727.45, SD \$346.12, and 95% CI \$2,717.95 to \$2,736.95, with half-width \$9.50. The same 5,100 samples are used in part (c).

Piedmont airline (c): booking limit

Compare limits 19 through 25 using common uniform draws across alternatives and $n = 5,100$. The native Excel Data Table changes Airline (c)!S2 and links to S8, the mean of K4:K5103. Exact expectations check the close alternatives independently.

Number of seats	22
Ticket price	150
Excess cost	500
Cancel. Prob.	0.1
Plane capacity	19
Mean profit	\$2,727.45
Samples	5100
Bookings	2727
19	\$2,721.15
20	\$2,763.75
21	\$2,754.97
22	\$2,727.45
23	\$2,705.04
24	\$2,691.28
25	\$2,685.82

Airline (c)!R2:S19.



Native Excel booking-limit chart.

Booking limit	Simulation mean	Exact expected profit
19	\$2,721.15	\$2,722.50
20	\$2,763.75	\$2,767.11
21	\$2,754.97	\$2,758.69
22	\$2,727.45	\$2,733.17
23	\$2,705.04	\$2,710.91
24	\$2,691.28	\$2,697.55
25	\$2,685.82	\$2,691.80

Recommend offering 20 tickets. Exact expected profit is \$2,767.11, compared with \$2,758.69 at 21 and \$2,733.17 at 22. Selling fewer than 19 tickets forgoes sales without reducing denied-boarding compensation. Limits above 25 are equivalent to 25 because demand never exceeds 25.

Food bank (a): stochastic model

Food types j are vegetables, rice, and cereal. Scenarios $s = 1, 2, 3$ have probabilities 0.40, 0.35, and 0.25. Each scenario jointly determines donations $a(s, j)$ and total demand $D(s, j)$, summed over the four organizations. Initial purchases are shared across scenarios.

Scenario	Probability	Canned vegetable	Rice	Cereal
1	40.0%	100	150	80
2	35.0%	200	100	150
3	25.0%	50	80	50

Scenario	Organization	Canned vegetable	Rice	Cereal
1	1	120	100	80
	2	100	120	100
	3	80	100	70
	4	100	80	90
	Total	400	400	340
2	1	140	120	100
	2	120	140	120
	3	100	120	90
	4	120	100	110
	Total	480	480	420
3	1	100	90	70
	2	80	100	80
	3	70	80	60
	4	90	70	80
	Total	340	340	290

Food bank!B4:F24: donations, organization demand, and scenario totals.

Variables: $x(j)$ is initial cases purchased; $E(s, j)$ is emergency cases purchased after observing scenario s . Both are nonnegative integers. Demand can be aggregated because there are no organization-specific allocation costs or restrictions.

Objective: Minimize $12x(V) + 10x(R) + 15x(C) + \sum_s p(s)[18E(s, V) + 16E(s, R) + 22E(s, C)]$.

Initial constraints: $x(V) + x(R) + x(C) \leq 700$; $12x(V) + 10x(R) + 15x(C) \leq 12000$.

Scenario constraints: $x(j) + a(s, j) + E(s, j) \geq D(s, j)$ for every scenario-food pair. Emergency spending uses separate funds and is outside the initial \$12,000 budget.

After donations are observed, food is allocated to satisfy every organization. An equivalent allocation model uses regular $y(s, i, j)$ and emergency $z(s, i, j)$, with $y + z = d(s, i, j)$, $\sum_i y \leq x + a$, and $\sum_i z = E$.

Food bank (b) and (c): optimal plan

Excel Solver with Simplex LP returned **280 vegetable cases, 180 rice cases, and 240 cereal cases**. Initial cost is **\$8,760**; warehouse use is 700 cases; initial budget remaining is \$3,240.

	Canned vegetable	Rice	Cereal
Purchase (x)	280	180	240
Emergency S1	20	70	20
Emergency S2	0	200	30
Emergency S3	10	80	0

Food bank!C28:F32. Initial and scenario emergency decisions are shown in cases.

Scenario	Vegetables	Rice	Cereal	Emergency cost
1	20	70	20	\$1,920
2	0	200	30	\$3,860
3	10	80	0	\$1,460

Expected emergency cost = $0.40 \times 1,920 + 0.35 \times 3,860 + 0.25 \times 1,460 = \$2,484$. Expected total cost = $8,760 + 2,484 = \mathbf{\$11,244}$. Purchase the initial quantities above and use emergency funds according to the realized scenario.

1. Storage capacity	700	≤	700
2. Budget	8760	≤	12,000

Scenario depend.

Scenario 1

Canned vegetables	400	≥	400
Rice	400	≥	400
Cereal	340	≥	340

Scenario 2

Canned vegetables	480	≥	480
Rice	480	≥	480
Cereal	420	≥	420

Scenario 3

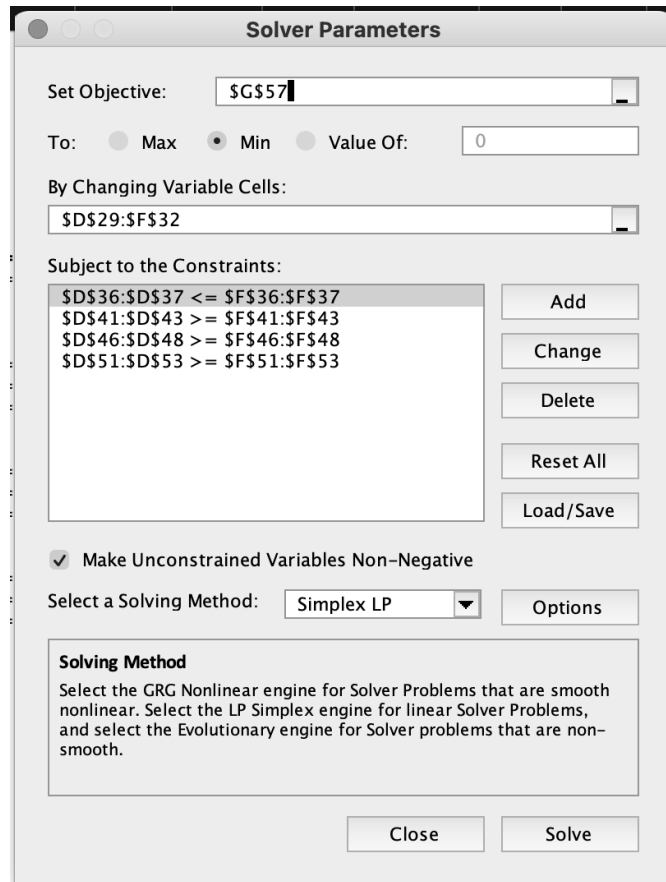
Canned vegetables	340	≥	340
Rice	340	≥	340
Cereal	290	≥	290

Food bank!C36:F53. All scenario demands are met; budget and capacity are satisfied. Scenario total costs are \$10,680, \$12,620, and \$10,220.

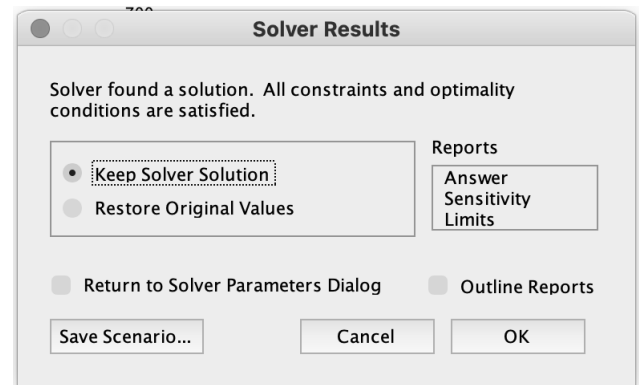
The LP optimum has whole-case decisions, so it is also feasible and optimal for the integer model. The earlier 210 / 250 / 240 purchase plan is an alternative optimum with the same \$11,244 expected cost. Warehouse capacity is binding; the initial budget is not.

Food bank: Excel Solver evidence

These are genuine screenshots of the Excel Solver dialogs used on the completed workbook. Parameters show the objective, decisions, grouped constraints, nonnegativity setting, and Simplex LP method. Results confirm feasibility and optimality.



Native Solver Parameters dialog.



Minimize G57.
 Changing cells: D29:F32.
 D36:D37 <= F36:F37.
 D41:D43 >= F41:F43.
 D46:D48 >= F46:F48.
 D51:D53 >= F51:F53.
 Nonnegative decisions.
 Simplex LP.

Saved objective calculation

OBJECTIVE FUNCTION			
Cost stage 1	\$8,760.00	Minimize cost	\$11,244.00
Scenario 1	\$1,920.00		
Scenario 2	\$3,860.00		
Scenario 3	\$1,460.00		

Food bank!C55:G60. G57 = D57 + SUMPRODUCT(D58:D60,C5:C7) = \$11,244.00. Solver's optimal solution was kept and the workbook saved.